Fine-mapping-based estimation of cis-heritability

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Outline:

1. What is cis-heritability?

2. How do we infer cis-heritability?

    GCTA

    SuSiE

3. Simulation

4. Result

5. Discussion

Why estimate cis-heritability of gene expression?

Gene expression reflects both local genetics and non-genetic/environmental factors.



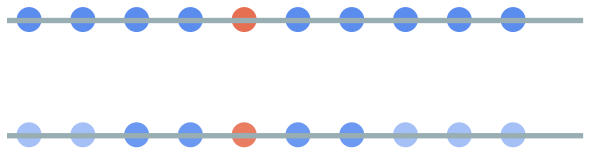
$$\text{cis}-h^2 = \frac{\text{Var}(\text{G_cis})}{[\,\text{Var}(\text{G_cis}) + \text{Var}(\text{E})\,]}$$

cis-h^2 tells us how much local inherited variation contributes to expression.

It helps prioritize genes and tissues for complex-trait interpretation and TWAS modeling.

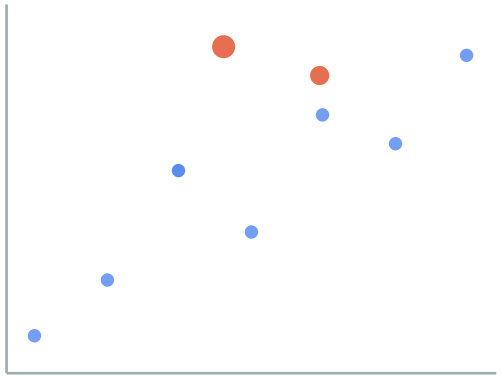
The challenge: LD makes marginal effects misleading

True architecture



one causal SNP,
many correlated neighbors

Association signals



LD can create several strong hits, even when only a subset is truly causal.

What can go wrong

Naïve sum

Adding marginal SNP effects can double-count shared signal.

Fine-mapping tries to assign probability to causal variants while accounting for LD.

What does GCTA try to do?

separate genetic signal from environmental noise using individual-level data.

Input for one gene

Expression vector y

One value per person: how strongly this gene is expressed after preprocessing.

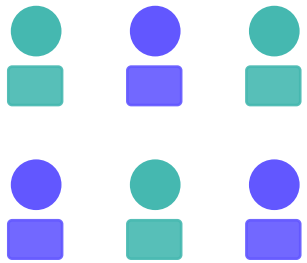
Genotype matrix Z

Rows are people; columns are cis-SNPs near the gene. Entries are standardized allele dosages.

Covariates X

Known non-genetic factors such as sex, age, batch, or genotype PCs.

Individual-level data



Same individuals measured for both genotype and expression

GCTA

+

REML

Output

cis-heritability

h^2_GE

How much expression variation can be attributed to local genetic variation?

GCTA treats the total genetic contribution as a random effect

Instead of estimating every SNP effect separately, GCTA estimates how much variance is explained by all cis-SNPs together.

$$y = X\beta + g + \varepsilon$$

y	$=$	$X\beta$	$+$	g	$+$	ε
expression		covariates		cis-genetic effect		environment / residual

GCTA asks: across people, how large is the variance of g compared with ε?

Fixed effects: Xβ

Remove predictable non-genetic effects first, such as batch, ancestry PCs, sex, or age.

Genetic random effect: g

Represents the combined contribution of all local cis-SNPs. We do not need to identify the exact causal SNPs.

Residual: ε

The remaining expression variation: environment, measurement noise, and unexplained biology.

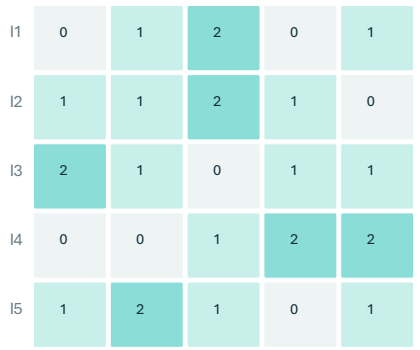
Model covariance: $Var(y) = A \cdot \sigma^2 g + I \cdot \sigma^2 e$

A = genetic relationship matrix; I = individual-specific residual noise

The genetic relationship matrix tells GCTA who is genetically similar

For a cis-gene analysis, the GRM is built from SNPs in the local region around that gene.

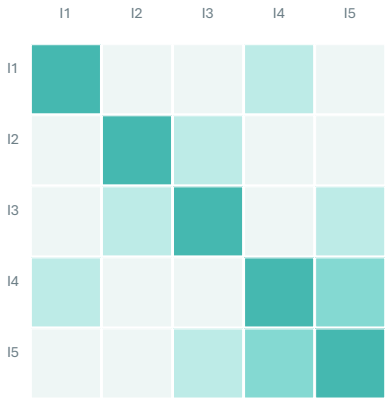
Standardized genotype matrix Z



people × local SNPs

$A = ZZ^T / p$

Genetic relationship matrix A



pairwise genetic similarity among individuals

How GCTA uses A

If two individuals are genetically similar in this cis-region, and their expression is also similar, GCTA attributes part of that pattern to $\sigma^2 g$.

similar genotypes → similar expression

GCTA uses genetic similarity across people to estimate the total cis-genetic variance.

How GCTA turns variance components into h^2

REML searches for the genetic and residual variance values that best explain the expression covariance across individuals.

GCTA estimation workflow



$$\text{cis-}h^2_GE = \frac{\sigma^2 g}{(\sigma^2 g + \sigma^2 e)}$$

genetic variance divided by total expression variance

GRM must be built from genotype data aligned to the same individuals and SNP region used for the expression model.

Why use SuSiE for cis-heritability?

SuSiE uses summary statistics and LD to infer a sparse genetic architecture, then we summarize the inferred effects as h^2.

Input for one gene

eQTL summary statistics

For each cis-SNP: marginal effect estimate, standard error, z-score, or p-value from association with gene expression.

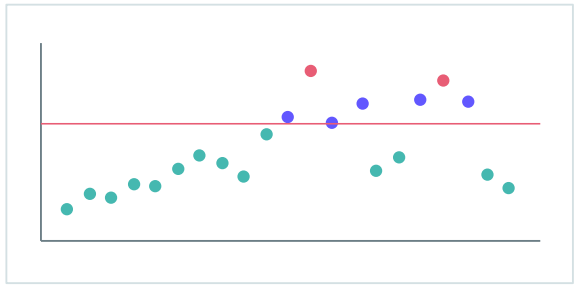
LD matrix R

Correlation between nearby SNPs. This tells SuSiE which association signals could be tagging the same causal variant.

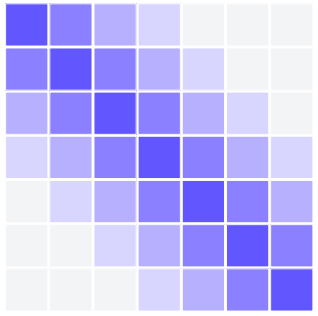
Sample size / settings

Needed to calibrate uncertainty and run fine-mapping with a chosen maximum number of effects L.

Summary-statistic fine-mapping



Many nearby SNPs can look associated because of LD



LD helps separate tagging from causal signal

SuSiE output

PIP — probability each SNP is causal

posterior mean β — estimated effect after accounting for LD

posterior SD — uncertainty of each effect

credible sets — groups of likely causal SNPs

then compute h^2 from posterior effects

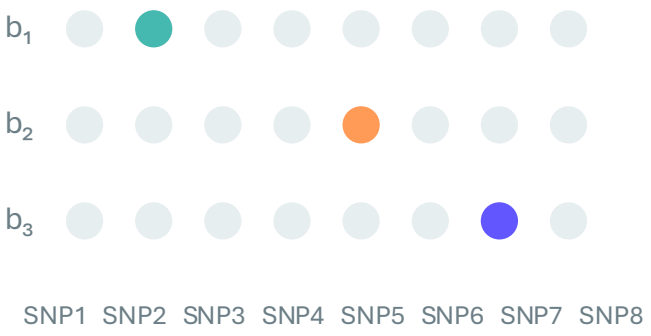
SuSiE models a sparse signal as a sum of single effects

A complex cis-region is decomposed into several simple signals, each allowed to select one likely causal SNP.

$$\beta = b_1 + b_2 + \cdots + bL$$

Each single-effect vector bk has at most one non-zero SNP effect.

Single-effect decomposition



What L means

L is the maximum number of independent association signals SuSiE is allowed to explain. It does not force exactly L causal SNPs.

Unused effects can shrink toward zero.

Main intuition

Nearby SNPs often move together because of LD. SuSiE assigns probability across these SNPs instead of choosing only one hard winner.

probabilistic fine-mapping

SuSiE says “this SNP has this posterior probability and this posterior effect distribution.”

How SuSiE updates effects using summary statistics and LD

The iterative Bayesian stepwise selection procedure repeatedly explains one signal while holding the others fixed.

Iterative Bayesian Stepwise Selection (IBSS)

1

Start with residual signal
Use summary statistics for the gene

2

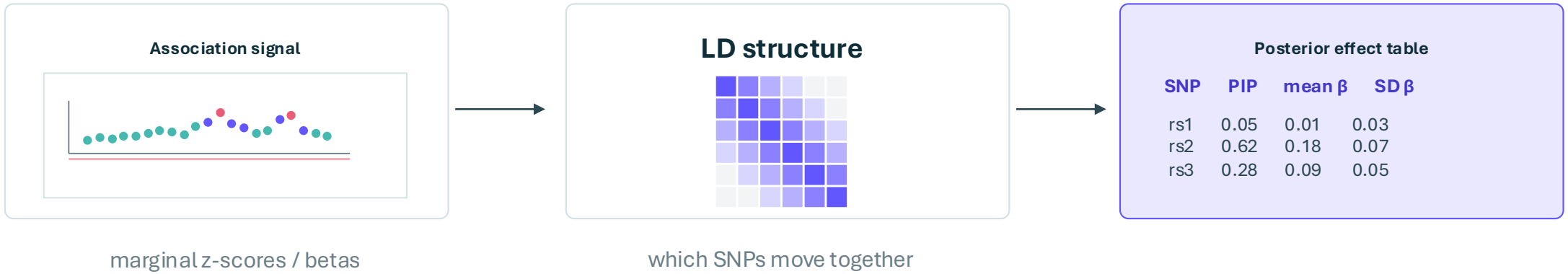
Update one effect
Which SNP best explains this residual?

3

Use LD to adjust
Avoid double-counting correlated SNPs

4

Repeat until stable
Return posterior means, SDs, and PIPs



marginal z-scores / betas

which SNPs move together

How we compute h^2 from SuSiE output

For standardized genotypes and expression, heritability is the sum of posterior second moments of SNP effects.

If genotype and expression are standardized:

$$h^2 \approx \Sigma j \; E[\beta j^2]$$

For each SNP, SuSiE gives a posterior effect distribution.
We add the expected squared effect across SNPs.

Using posterior mean and SD:

$$E[\beta j^2] = \mu j^2 + \sigma j^2$$

This keeps both the estimated effect size and posterior uncertainty.

In our simulation code

```
su_hsq = sum(POSTERIOR² + POSTERIOR_SD²)
```

same goal as GCTA, different input data

It is SuSiE's posterior estimate of the fraction of gene-expression variance explained by local genetic effects.
In our project, we compare this summary-statistic estimate against true simulated h^2 and GCTA.

Two estimators in this project

GCTA is the individual-level benchmark; SuSiE is the summary-statistic fine-mapping estimator.

GCTA / REML benchmark

Input: simulated genotype matrix Z and expression y.

Build GRM: $A = ZZ^T / p$.

Fit mixed model: $y = X\beta + g + \varepsilon$.

Output: h^2 estimate, SE, p-value.

$$h^2_GCTA = \sigma^2_g / (\sigma^2_g + \sigma^2_e)$$

SuSiE summary-statistic estimator

Input: single-SNP regression summary statistics and LD matrix.

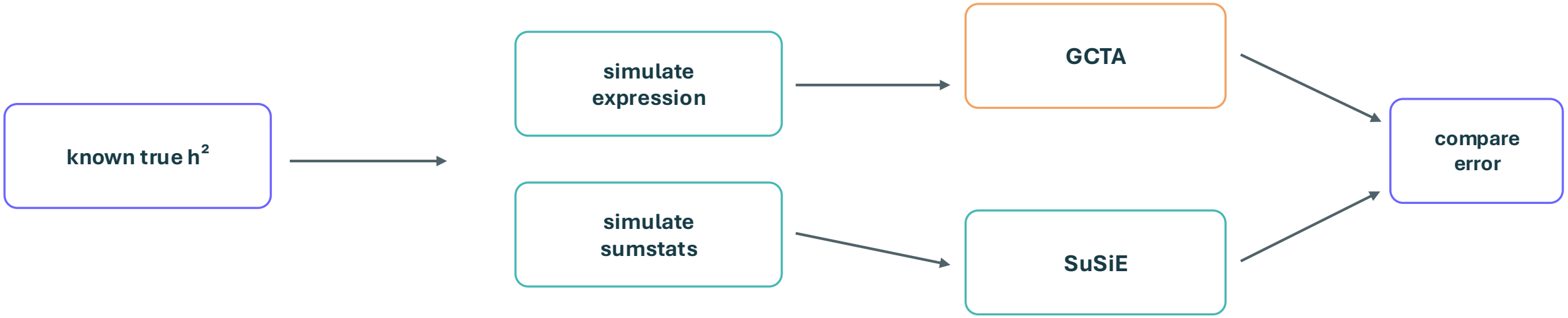
Fine-map to infer posterior effects.

Estimate heritability by summing posterior second moments across SNPs.

$$h^2_SuSiE = \Sigma_j (\mu_j^2 + \sigma_j^2)$$

Main question for the simulation study

When does fine-mapping-based cis-heritability agree with GCTA, and when does it fail?



Reliability

Are estimates close to the truth and stable across replicates?

Architecture

Which genetic settings create over- or under-estimation?

Simulation workflow

The code generates expression data with known truth, then benchmarks GCTA and SuSiE.

1 → 2 → 3 → 4 → 5 → 6

Choose region

random chromosome + 1 Mb window from 1000G EUR PLINK files

Build LD

standardize genotype; compute LD; regularize and Cholesky-decompose

Simulate Z

sample standardized genotypes from LD approximation

Simulate effects

select causal SNPs; assign effect sizes; scale to target h^2

Simulate expression

latent genetic value + noise; standardize phenotype

Run estimators

GCTA from GRM; SuSiE from sumstats + LD

Targeted grid: ncausal × h^2 × architecture × replicate

Step 1: create a realistic local LD structure

Use real genotype regions as the source of SNP correlation, then simulate standardized genotypes.



Standardization

Center and scale each SNP so the simulated genotype matrix has unit variance per SNP.

Regularization

Add a small diagonal term so the LD matrix is positive definite for Cholesky sampling.

Output

A local correlation structure that can generate realistic correlated genotypes.

Step 2: simulate causal effects and expression

The simulation fixes the true cis-heritability, then adds environmental noise.

$$Z = L \cdot \text{Normal}(0, I) \longrightarrow g = Z\beta \longrightarrow y = g + \varepsilon$$

Causal SNPs

Pick ncausal SNPs within the local region. Baseline code uses ncausal = 1, 2, 5, 10; extensions include up to 20.

Effect scaling

Draw β, then rescale so $\beta^T R \beta$ equals the target genetic variance for the selected h^2.

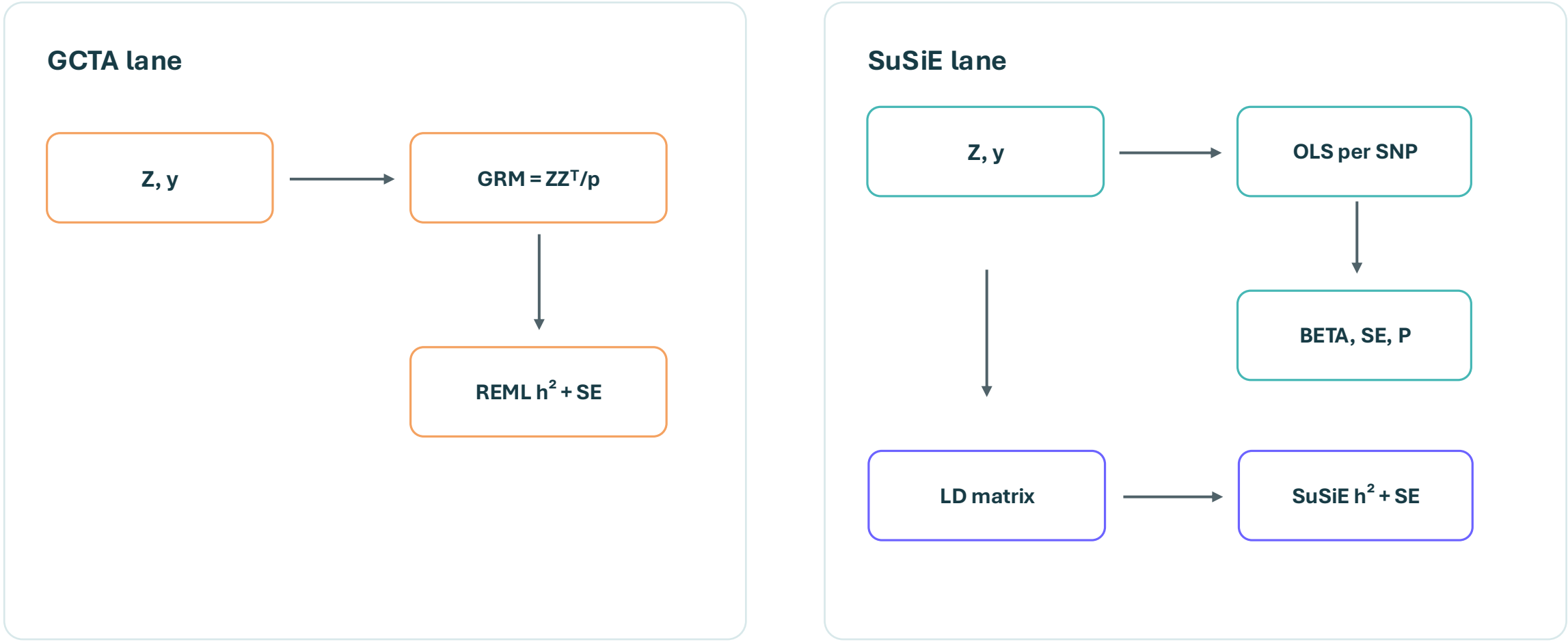
Noise calibration

Choose noise variance so final standardized expression has the target cis-heritability.

$$\text{Var}(g) \,/\, \text{Var}(y) = \text{target cis-}h^2$$

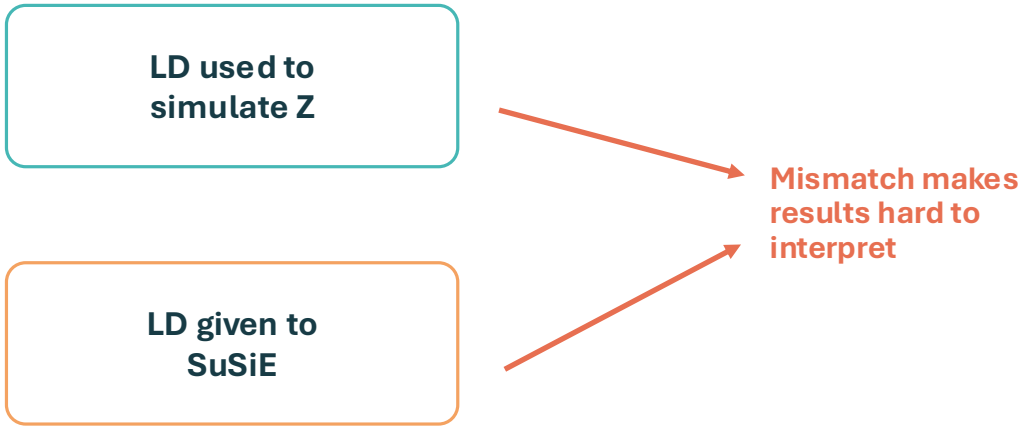
Step 3: give each method the data it expects

GCTA uses individual-level simulated data; SuSiE uses summary statistics plus LD.

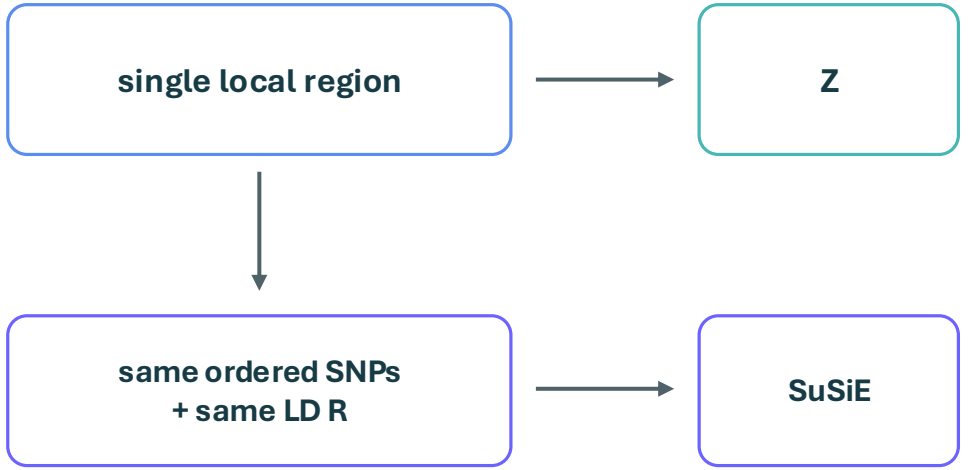


Key method principle: LD must be aligned across the pipeline

Previous risk



Target design



Inference LD should match the covariance structure of the simulated genotypes.

What the code is designed to stress-test

Each script isolates one genetic architecture factor while keeping the benchmark logic similar.

Baseline grid

ncausal = 1, 2, 5, 10
h^2 = 0.05–0.8
N = 300

Effect-size distribution

normal effects
fixed-size effects
heavy-tailed Pareto effects

Allele frequency

causal SNPs sampled from MAF bins to test rare vs common variants

LD architecture

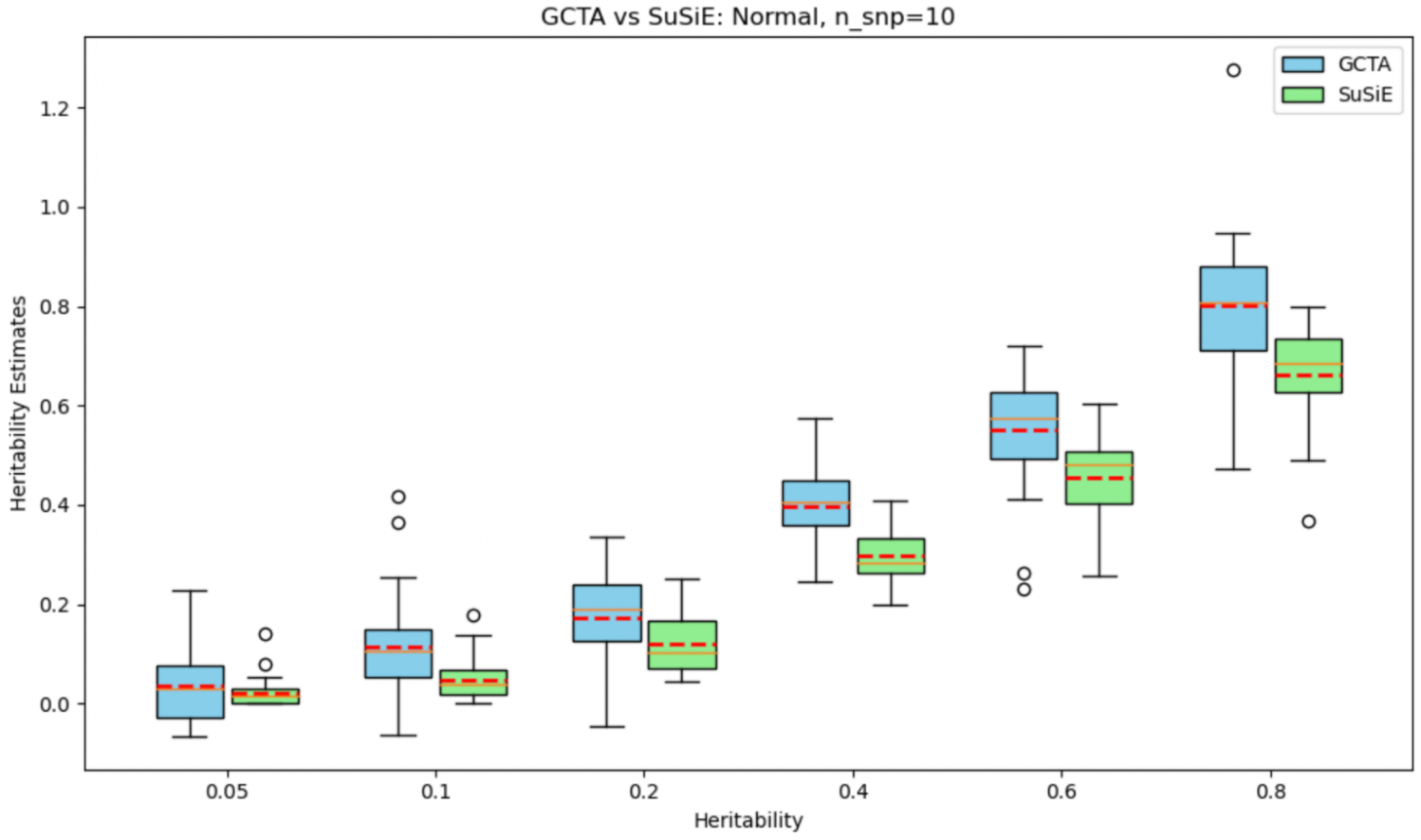
high-LD causal SNPs
low-LD region filtering
LD summary by r^2 threshold

Output metrics

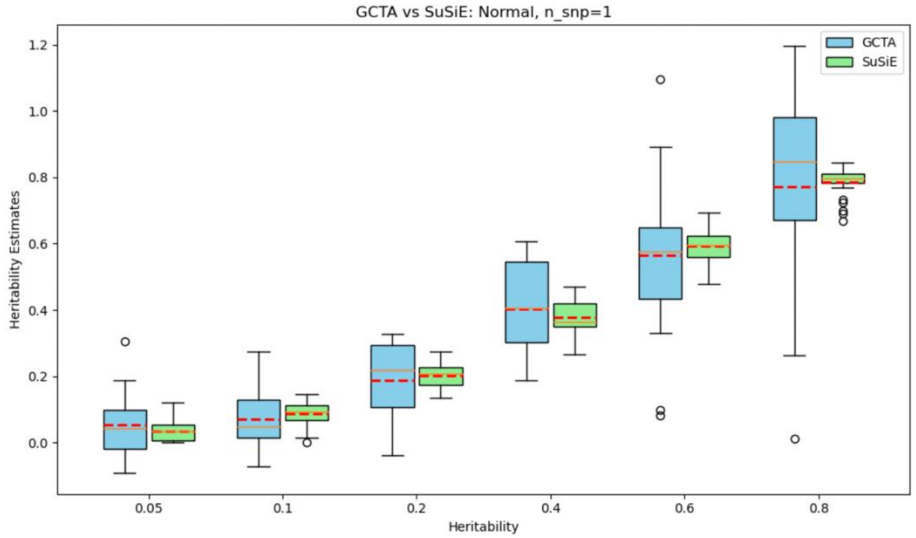
GCTA h^2, SE, p
SuSiE h^2, analytic SE
method error vs truth

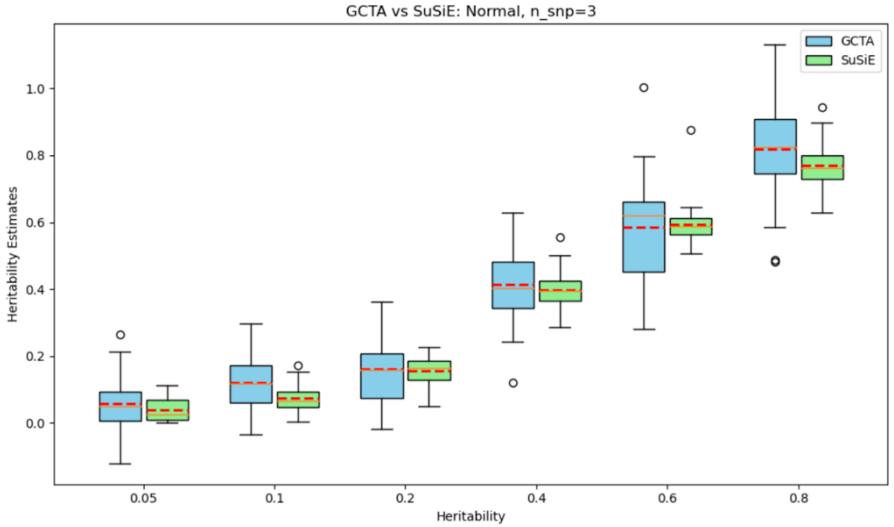
Goal: identify conditions that cause bias or instability

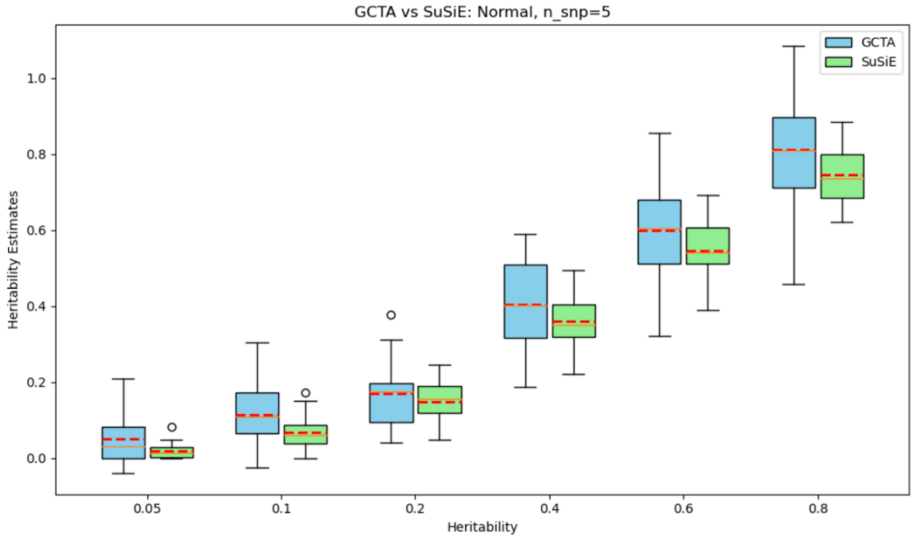
Result : Random Selected Region

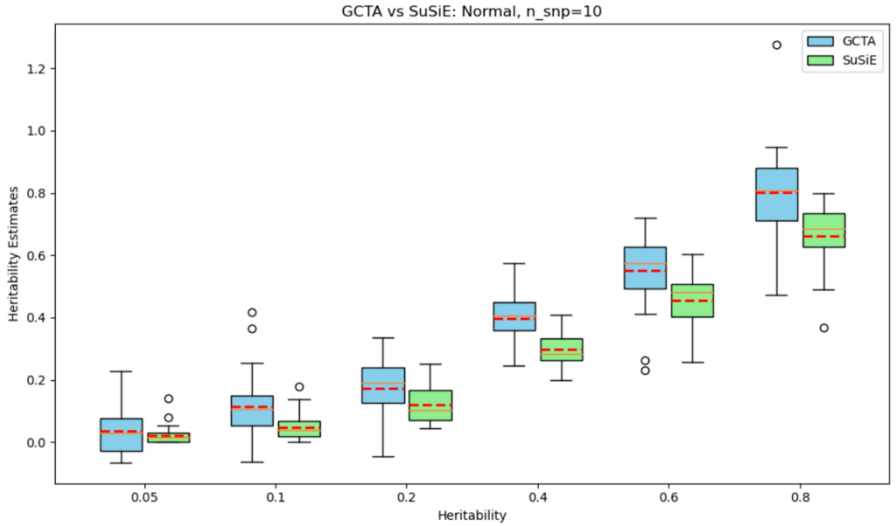


Result : Random Selected Region



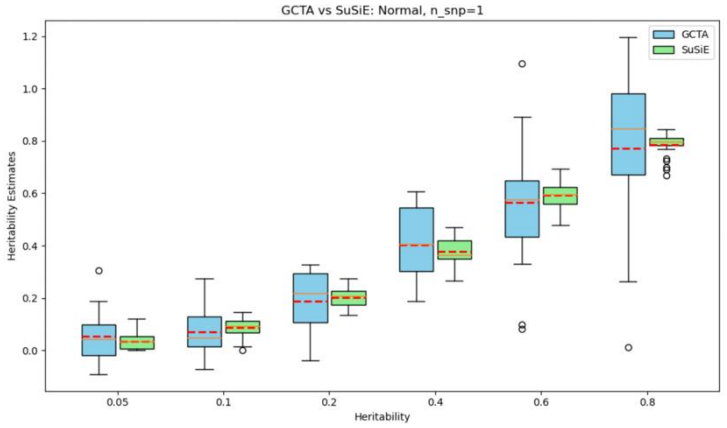




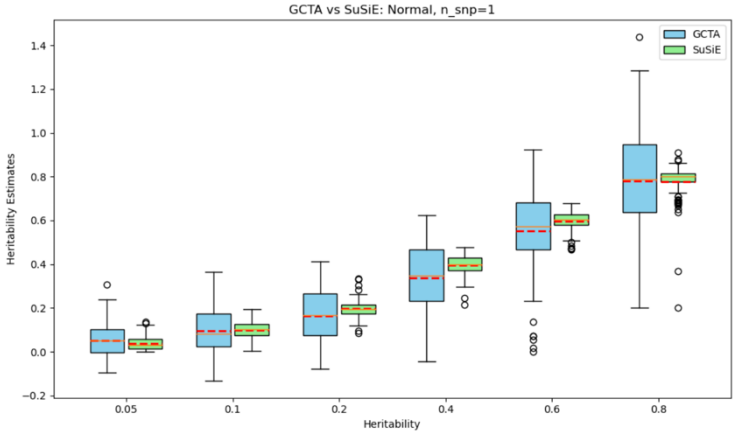


SuSiE generally outperforms GCTA when **h^2 is low** (0.05–0.2), when the number of **causal SNPs is small** (1–3), or both.

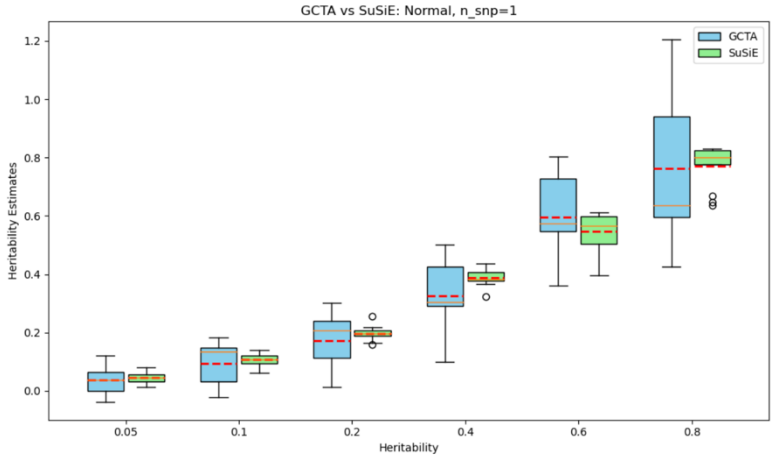
Result : Different Sample Sizes (n_snp = 1/3)



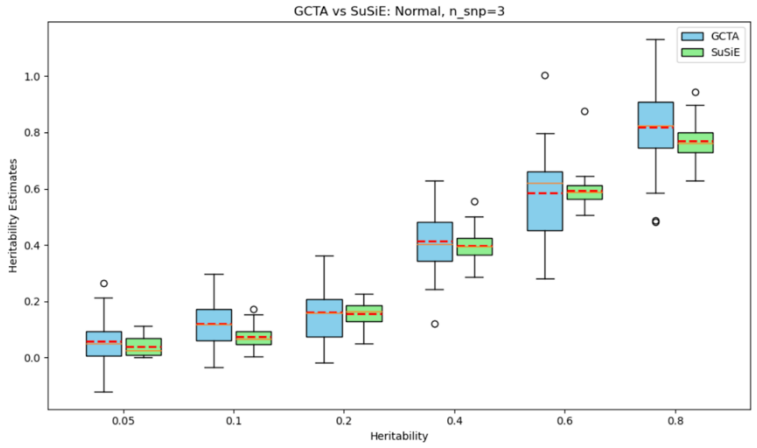
n=100



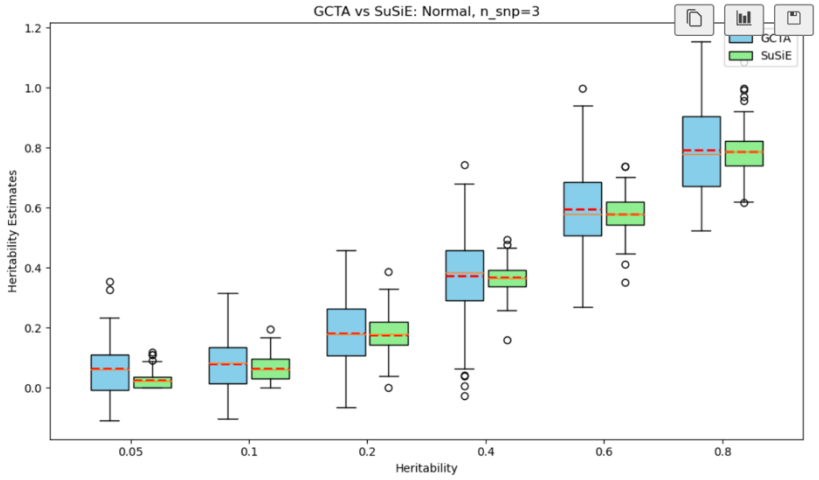
n=300



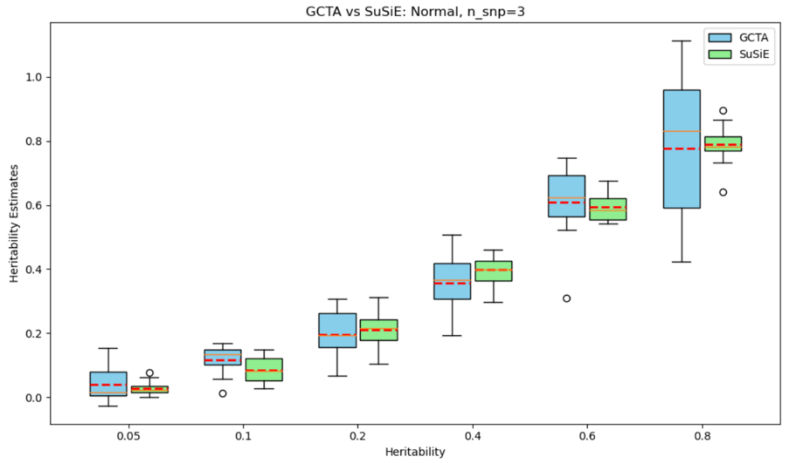
n=500



n=100

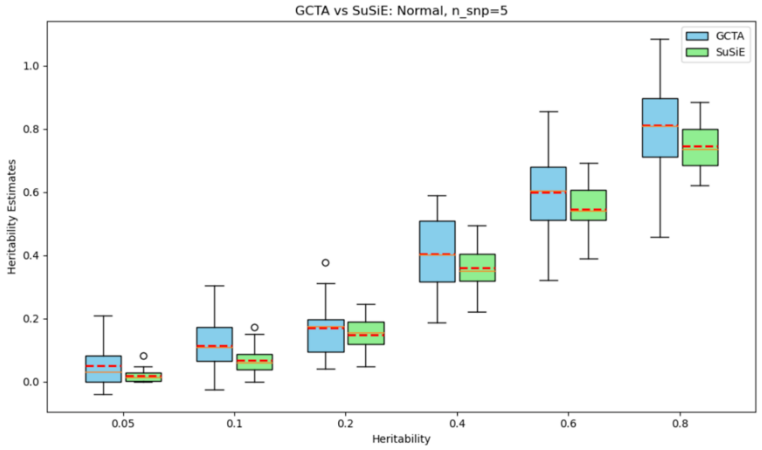


n=300

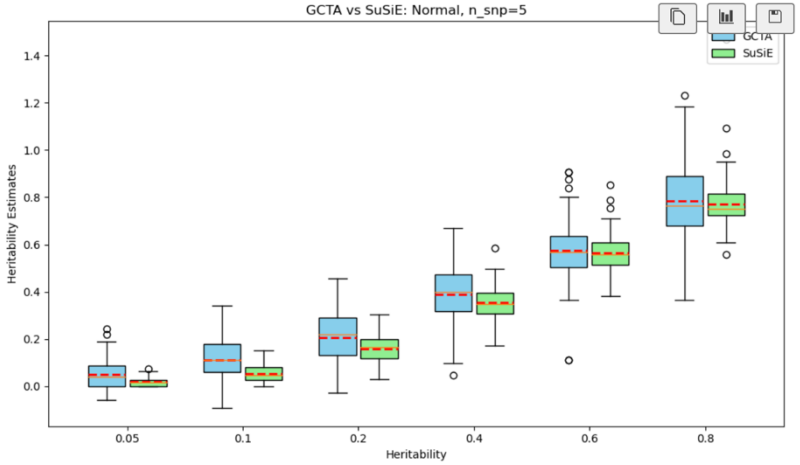


n=500

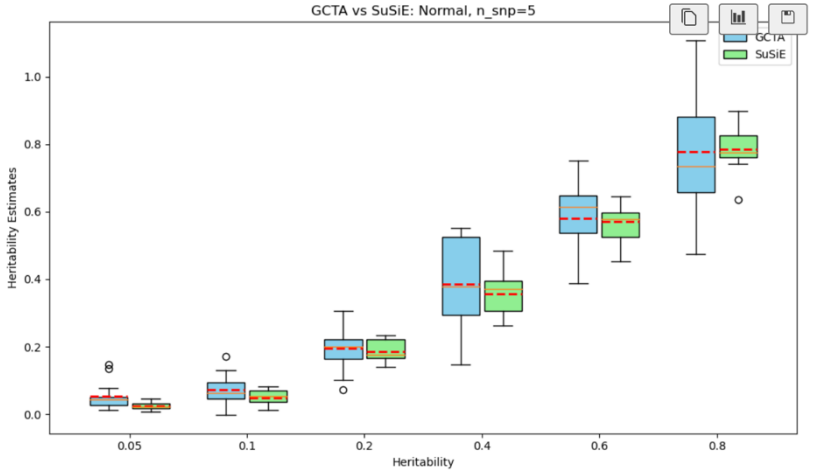
Result : Different Sample Sizes (n_snp = 5/10)



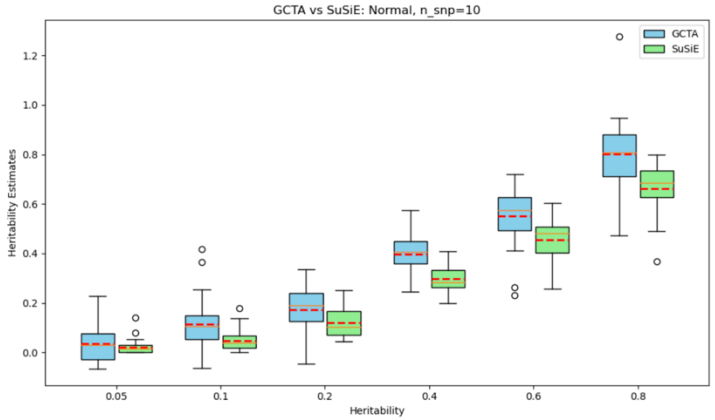
n=100



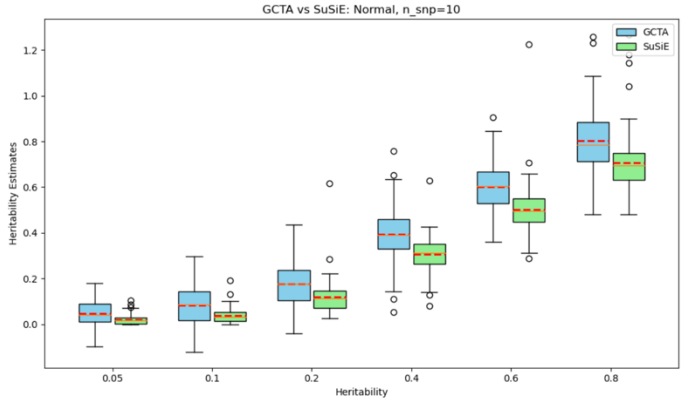
n=300



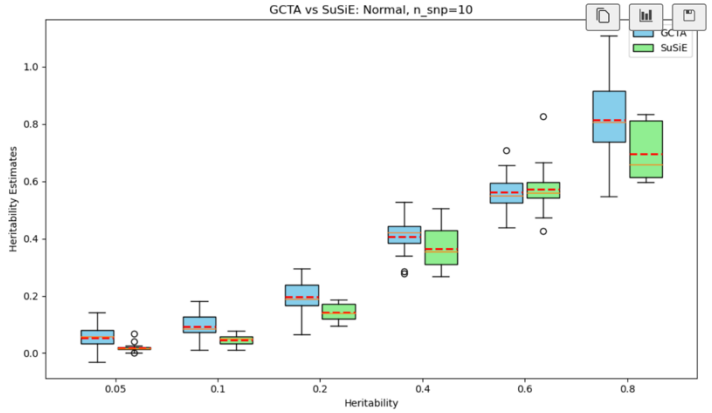
n=500



n=100



n=300

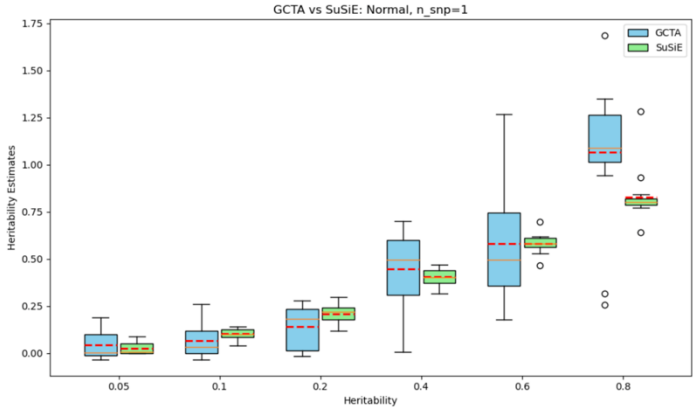


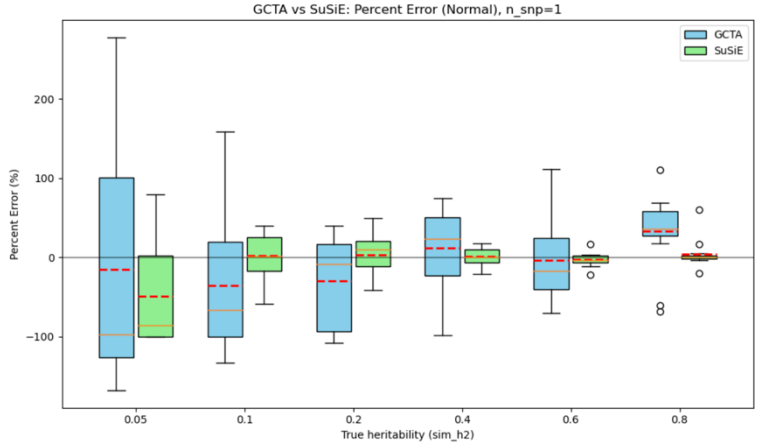
n=500

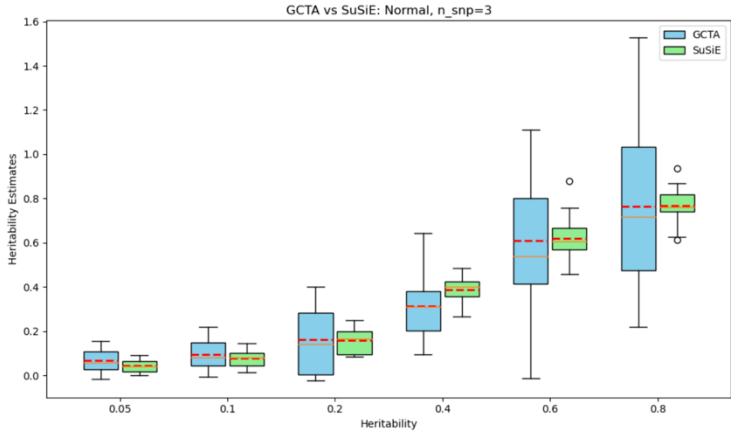
Sample size does not substantially affect the overall predictions of the two methods, but GCTA may show a slightly larger standard deviation when n is large.

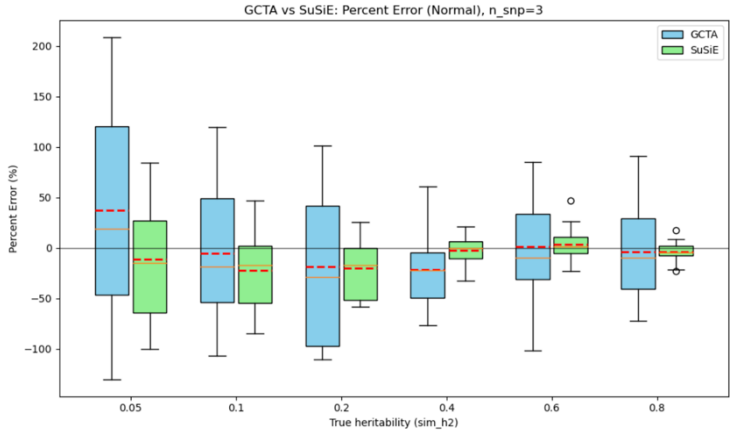
Result : High LD

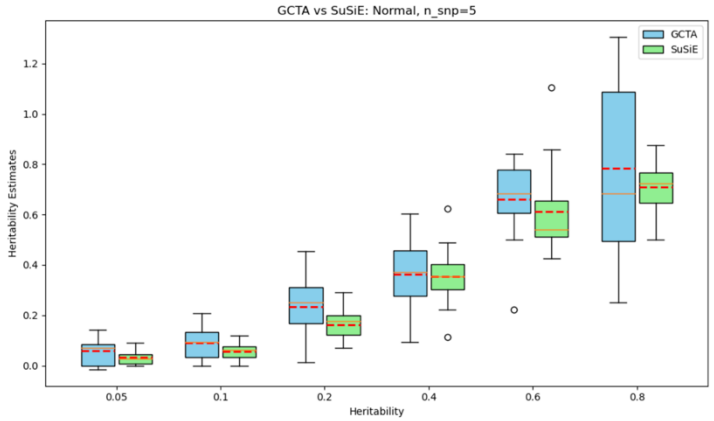
Under high-LD scenarios, SuSiE tends to perform better than in normal LD settings and often outperforms GCTA, even at high h^2 values.

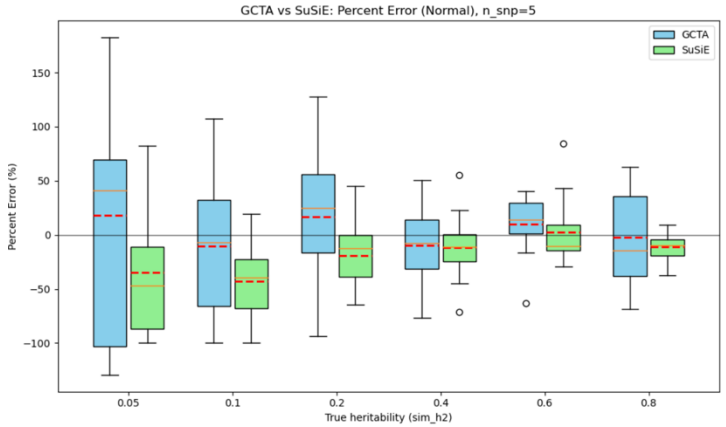








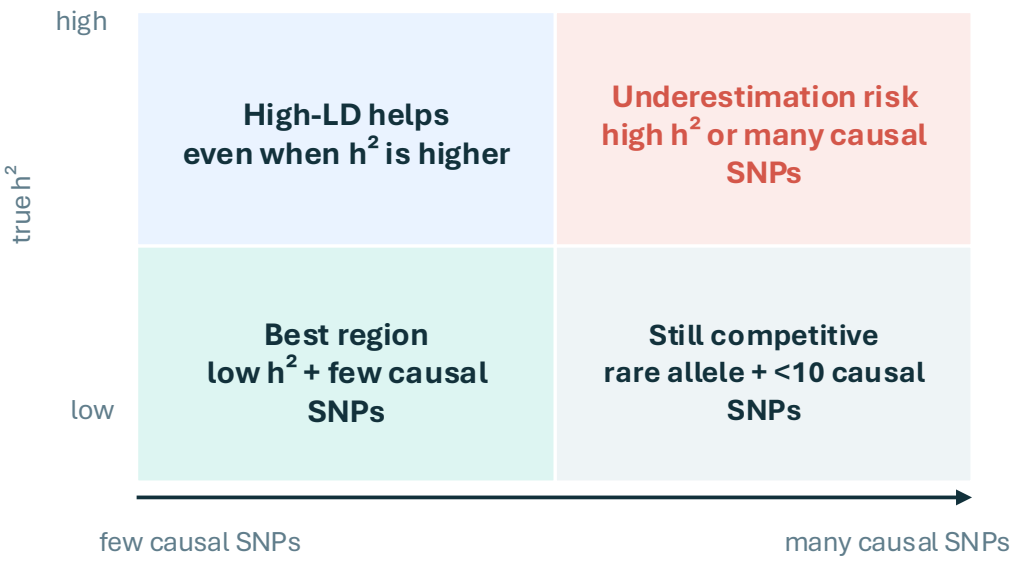




Where does SuSiE outperform GCTA?

Across the simulations, SuSiE performs best when the cis-genetic architecture is sparse or LD provides useful information.

Performance pattern across architectures



Main findings

- SuSiE generally outperforms GCTA when true h^2 is low (0.05–0.2).
- SuSiE is strongest when the number of causal SNPs is small (1–3), especially when combined with low h^2.
- Under high-LD scenarios, SuSiE often improves over the normal-LD setting and frequently beats GCTA, even at high h^2.
- In rare allele-frequency simulations, SuSiE is slightly better when the number of causal SNPs is below 10.

Caution: SuSiE starts to underestimate when h^2 is high or when the causal SNPs becomes dense.

Discussion / Future Work

- Test whether the same patterns appear in real GTEx data
- Check if SuSiE performs better for genes with low h^2, sparse causal SNPs, or high local LD
- Compare SuSiE and GCTA with a third method: HEELS
- HEELS also uses summary statistics and LD, making it a useful benchmark for SuSiE
- Goal: identify when SuSiE-based cis-heritability estimation is reliable, biased, or method-dependent

Thank you!

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